

Practical task

1. Install Nexus
2. Configure admin user
3. Configure proxy repo
 - a. For Maven: Create setting.xml, maven should use it as mirror
 - b. For Gradle: Define proxy repo in gradle settings
4. Upload spring-petclinic Artifact to Nexus
 - a. For Maven: Use the deploy plugin in your pom.xml. You should be able to upload your artifacts to Nexus using Maven.
 - b. For Gradle: Define artifact upload as a separate Gradle task
5. Search artifacts in Nexus

1. Install Nexus

Install Nexus Repository Manager

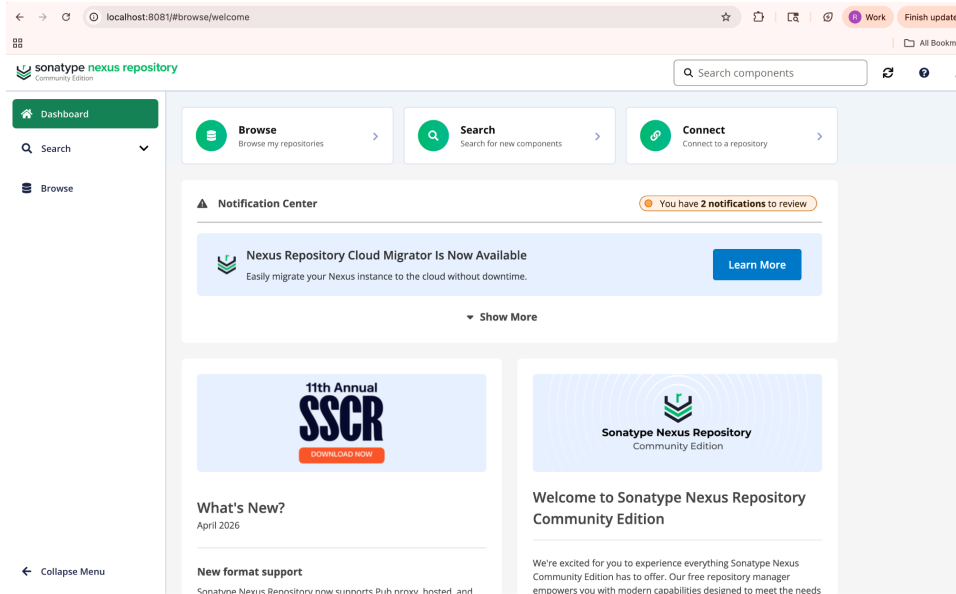
Nexus Repository Manager was installed and started using Docker.

Command Used

```
dsarrafgriiddynamics.com@C19548 Assignments % docker run -d \ --name nexus \ -p 8081:8081 \ sonatype/nexus3
```

Verification

Nexus UI was successfully accessed through: <http://localhost:8081>



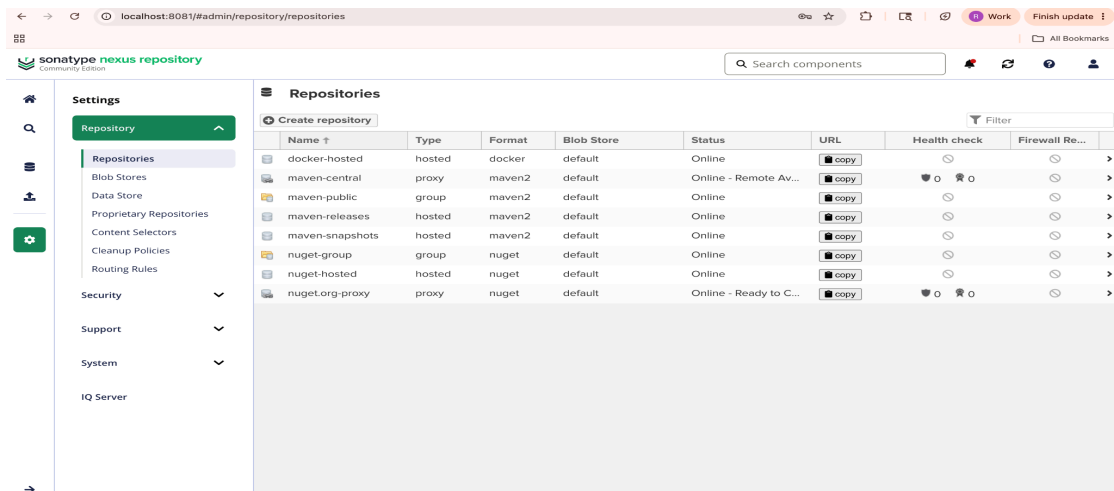
2. Configure Admin User

1. Retrieved the initial admin password from the Nexus container.
2. Logged in using the default admin account.
3. Changed the default password.
4. Completed the initial Nexus setup wizard.

```
dsarrafriddynamics.com@C19548 Assignments % docker exec -it nexus sh
/opt/sonatype $ cat /nexus-data/admin.password
```

3. Configure Maven Repositories

The default Nexus repositories were already available:



These repositories were used for dependency caching and artifact storage.

a. Configure Maven settings.xml

A Maven settings file was created to configure Nexus as a mirror repository.

Location: `~/.m2/settings.xml`

Configuration:

```
<settings>

  <mirrors>

    <mirror>

      <id>nexus</id>

      <mirrorOf>*</mirrorOf>

      <url>http://localhost:8081/repository/maven-public/</url>

    </mirror>

  </mirrors>

  <servers>

    <server>

      <id>nexus</id>

      <username>admin</username>

      <password>*****</password>

    </server>

  </servers>

</settings>
```

Result:

- Redirects all Maven dependency downloads through Nexus.
- Uses Nexus as a mirror for Maven Central.
- Provides authentication credentials for artifact deployment.

b. Gradle Configuration

The Spring Petclinic project already contained Gradle wrapper files:

build.gradle

gradlew

Gradlew.bat

Settings.gradle

```
dsarrafriddynamics.com@C19548 spring-petclinic % ls
Dockerfile.basic  gradle      pom.xml
Dockerfile.multi  gradlew     settings.gradle
LICENSE.txt       gradlew.bat src
README.md        k8s        target
build.gradle      mvnw
docker-compose.yml mvnw.cmd
(.venv) dsarrafriddynamics.com@C19548 spring-petclinic % cat build.gradle
plugins {
    id 'java'
    id 'checkstyle'
    id 'org.springframework.boot' version '4.0.3'
    id 'io.spring.dependency-management' version '1.1.7'
    id 'org.graalvm.buildtools.native' version '0.11.5'
    id 'org.cyclonedx.bom' version '3.2.0'
    id 'io.spring.javaformat' version '0.0.47'
    id "io.spring.nohttp" version "0.0.11"
}

gradle.startParameter.excludedTaskNames += [ "checkFormatAot", "checkFormatAotTest" ]

group = 'org.springframework.samples'
version = '4.0.0-SNAPSHOT'

java {
    toolchain {
        languageVersion = JavaLanguageVersion.of(17)
    }
}

repositories {
    mavenCentral()
}

ext.checkstyleVersion = "12.3.1"
ext.springJavaformatCheckstyleVersion = "0.0.47"
ext.webjarsLocatorLiteVersion = "1.1.3"
ext.webjarsFontawesomeVersion = "4.7.0"
ext.webjarsBootstrapVersion = "5.3.8"

dependencies {
    implementation 'org.springframework.boot:spring-boot-starter-cache'
    implementation 'org.springframework.boot:spring-boot-starter-data-jpa'
    implementation 'org.springframework.boot:spring-boot-starter-thymeleaf'
    implementation 'org.springframework.boot:spring-boot-starter-webmvc'
    implementation 'org.springframework.boot:spring-boot-starter-validation'
    implementation 'javax.cache:cache-api'
```

```

implementation 'jakarta.xml.bind:jakarta.xml.bind-api'
runtimeOnly 'org.springframework.boot:spring-boot-starter-actuator'
runtimeOnly "org.webjars:webjars-locator-lite:${webjarsLocatorLiteVersion}"
runtimeOnly "org.webjars.npm:bootstrap:${webjarsBootstrapVersion}"
runtimeOnly "org.webjars.npm:font-awesome:${webjarsFontawesomeVersion}"
runtimeOnly 'com.github.ben-manes.caffeine:caffeine'
runtimeOnly 'com.h2database:h2'
runtimeOnly 'com.mysql:mysql-connector-j'
runtimeOnly 'org.postgresql:postgresql'
developmentOnly 'org.springframework.boot:spring-boot-devtools'
testImplementation 'org.springframework.boot:spring-boot-starter-data-jpa-test'
testImplementation 'org.springframework.boot:spring-boot-starter-restclient-test'
testImplementation 'org.springframework.boot:spring-boot-starter-webmvc-test'
testImplementation 'org.springframework.boot:spring-boot-testcontainers'
testImplementation 'org.springframework.boot:spring-boot-docker-compose'
testImplementation 'org.testcontainers:testcontainers-junit-jupiter'
testImplementation 'org.testcontainers:testcontainers-mysql'
checkstyle "io.spring.javaformat:spring-javaformat-checkstyle:${springJavaformatCheckstyleVersion}"
checkstyle "com.puppycrawl.tools:checkstyle:${checkstyleVersion}"
}

tasks.named('test') {
    useJUnitPlatform()
}

tasks.named("cyclonedxDirectBom").configure {
    dependsOn("compileJava")
    skipConfigs = [".*[Tt]est.*"]
}

checkstyle {
    configDirectory = project.file('src/checkstyle')
    configFile = file('src/checkstyle/nohttp-checkstyle.xml')
}

checkstyleNohttp {
    configDirectory = project.file('src/checkstyle')
    configFile = file('src/checkstyle/nohttp-checkstyle.xml')
}

tasks.named("formatMain").configure { dependsOn("checkstyleMain") }
tasks.named("formatMain").configure { dependsOn("checkstyleNohttp") }

tasks.named("formatTest").configure { dependsOn("checkstyleTest") }
tasks.named("formatTest").configure { dependsOn("checkstyleNohttp") }

checkstyleAot.enabled = false
checkstyleAotTest.enabled = false

checkFormatAot.enabled = false
checkFormatAotTest.enabled = false

formatAot.enabled = false
formatAotTest.enabled = false%

```

Add Maven Publish Plugin

```
plugins {  
    id 'maven-publish'  
}
```

Configure Nexus Proxy Repository

```
repositories {  
    maven {  
        url "http://localhost:8081/repository/maven-public/"  
        allowInsecureProtocol = true  
    }  
}
```

Purpose

- Gradle downloads dependencies through Nexus instead of directly from Maven Central.
- Enables repository caching and centralized dependency management.

Configure Artifact Publishing

Added the following block to `build.gradle`:

```
publishing {  
  
    publications {  
  
        mavenJava(MavenPublication) {  
  
            from components.java  
  
        }  
  
    }  
  
    repositories {  
  
        maven {  
  
            url = uri("http://localhost:8081/repository/maven-snapshots/")  
  
            allowInsecureProtocol = true  
  
            credentials {  
  
                username = "admin"  
  
            }  
  
        }  
  
    }  
}
```

```
password = "*****"
```

```
}
```

```
}
```

```
}
```

```
}
```

Verify Publish Tasks : `./gradlew tasks --all | grep publish`

```
dsarrafriddynamics.com@C19548 spring-petclinic % ./gradlew tasks --all | grep publish
publish - Publishes all publications produced by this project.
publishAllPublicationsToMavenRepository - Publishes all Maven publications produced by this project
to the maven repository.
publishMavenJavaPublicationToMavenLocal - Publishes Maven publication 'mavenJava' to the local
Maven repository.
publishMavenJavaPublicationToMavenRepository - Publishes Maven publication 'mavenJava' to Maven
repository 'maven'.
publishToMavenLocal - Publishes all Maven publications produced by this project to the local Maven
cache.
```

4. Configure Spring Petclinic Deployment

The Spring Petclinic project's pom.xml file was updated with distribution management configuration.

```
<distributionManagement>
```

```
  <snapshotRepository>
```

```
    <id>nexus</id>
```

```
    <name>Nexus Snapshots</name>
```

```
    <url>http://localhost:8081/repository/maven-snapshots</url>
```

```
  </snapshotRepository>
```

```
</distributionManagement>
```

Note:

The application version was:

4.0.0-SNAPSHOT

Therefore deployment was performed to the maven-snapshots repository instead of maven-releases.

a. Build Spring Petclinic

The application was built using Maven.

Command

```
./mvnw clean package -DskipTests
```

Result:

- Application compiled successfully.
- JAR artifact was generated.

```
dsarrafriddynamics.com@C19548 spring-petclinic % ./mvnw clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.springframework.samples:spring-petclinic >-----
[INFO] Building petclinic 4.0.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ spring-petclinic ---
[INFO] Deleting /Users/dsarrafr/Downloads/Assignments/spring-petclinic/target
[INFO]
[INFO] --- enforcer:3.6.2:enforce (enforce-java) @ spring-petclinic ---
[INFO] Rule 0: org.apache.maven.enforcer.rules.version.RequireJavaVersion passed
[INFO]
[INFO] --- spring-javaformat:0.0.47:validate (default) @ spring-petclinic ---
[INFO]
[INFO] --- checkstyle:3.6.0:check (nohttp-checkstyle-validation) @ spring-petclinic ---
[INFO] You have 0 Checkstyle violations.
[INFO]
[INFO] --- jacoco:0.8.14:prepare-agent (default) @ spring-petclinic ---
[INFO] argLine set to
-javaagent:/Users/dsarrafr/.m2/repository/org/jacoco/org.jacoco.agent/0.8.14/org.jacoco.agent-0.8.14-runtime.jar
=destfile=/Users/dsarrafr/Downloads/Assignments/spring-petclinic/target/jacoco.exec
[INFO]
[INFO] --- git-commit-id:9.0.2:revision (default) @ spring-petclinic ---
[INFO] dotGitDirectory '/Users/dsarrafr/Downloads/Assignments/.git'
[INFO] Collected git.build.user.name with value dsarrafr
[INFO] Collected git.build.user.email with value dsarrafr@griddynamics.com
[INFO] Collected git.branch with value main
[INFO] Collected git.commit.id.describe with value d2b637c-dirty
[INFO] Collected git.commit.id.describe-short with value d2b637c-dirty
[INFO] Collected git.commit.id with value d2b637c687717cfeeb7b197a391ba6c087a53243
[INFO] Collected git.commit.id.abbrev with value d2b637c
[INFO] Collected git.dirty with value true
[INFO] Collected git.commit.user.name with value dsarrafr
[INFO] Collected git.commit.user.email with value dsarrafr@griddynamics.com
[INFO] Collected git.commit.message.full with value docker practical task
[INFO] Collected git.commit.message.short with value docker practical task
[INFO] Collected git.commit.time with value 2026-05-01T11:51:43+05:30
[INFO] Collected git.commit.author.time with value 2026-05-01T11:51:43+05:30
```



```
[INFO] Collected git.commit.committer.time with value 2026-05-01T11:51:43+05:30
[INFO] Collected git.remote.origin.url with value https://github.com/dsarraaf-png/Assignments.git
[INFO] Collected git.tags with value
[INFO] Collected git.tag with value
[INFO] Collected git.closest.tag.name with value
[INFO] Collected git.closest.tag.commit.count with value
[INFO] Collected git.total.commit.count with value 29
[INFO] Collected git.local.branch.ahead with value 0
[INFO] Collected git.local.branch.behind with value 0
[INFO] Collected git.build.time with value 2024-11-28T20:07:52+05:30
[INFO] Collected git.build.version with value 4.0.0-SNAPSHOT
[INFO] Collected git.build.host with value C19536
[INFO] including property 'git.tags' in results
[INFO] including property 'git.build.version' in results
[INFO] including property 'git.branch' in results
[INFO] including property 'git.build.host' in results
[INFO] including property 'git.commit.id' in results
[INFO] including property 'git.commit.user.email' in results
[INFO] including property 'git.local.branch.behind' in results
[INFO] including property 'git.commit.author.time' in results
[INFO] including property 'git.build.user.name' in results
[INFO] including property 'git.dirty' in results
[INFO] including property 'git.closest.tag.commit.count' in results
[INFO] including property 'git.commit.user.name' in results
[INFO] including property 'git.commit.id.abbrev' in results
[INFO] including property 'git.commit.id.describe-short' in results
[INFO] including property 'git.total.commit.count' in results
[INFO] including property 'git.commit.id.describe' in results
[INFO] including property 'git.build.user.email' in results
[INFO] including property 'git.commit.message.short' in results
[INFO] including property 'git.commit.committer.time' in results
[INFO] including property 'git.tag' in results
[INFO] including property 'git.closest.tag.name' in results
[INFO] including property 'git.local.branch.ahead' in results
[INFO] including property 'git.commit.time' in results
[INFO] including property 'git.build.time' in results
[INFO] including property 'git.commit.message.full' in results
[INFO] including property 'git.remote.origin.url' in results
[INFO] Writing properties file
[INFO] [/Users/dsarraaf/Downloads/Assignments/spring-petclinic/target/classes/git.properties] (for project petclinic)...
[INFO]
[INFO] --- spring-boot:4.0.3:build-info (default) @ spring-petclinic ---
[INFO]
[INFO] --- cyclonedx:2.9.1:makeAggregateBom (default) @ spring-petclinic ---
[INFO] CycloneDX: Resolving Dependencies
[INFO] CycloneDX: Creating BOM version 1.6 with 108 component(s)
[INFO] CycloneDX: Writing and validating BOM (JSON):
[INFO] /Users/dsarraaf/Downloads/Assignments/spring-petclinic/target/classes/META-INF/sbom/application.cdx.json
[WARNING] Unknown keyword meta:enum - you should define your own Meta Schema. If the keyword is irrelevant
for validation, just use a NonValidationKeyword or if it should generate annotations AnnotationKeyword
[WARNING] Unknown keyword deprecated - you should define your own Meta Schema. If the keyword is irrelevant
for validation, just use a NonValidationKeyword or if it should generate annotations AnnotationKeyword
[INFO] attaching as spring-petclinic-4.0.0-SNAPSHOT-cyclonedx.json
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ spring-petclinic ---
[INFO] Copying 3 resources from src/main/resources to target/classes
[INFO] Copying 45 resources from src/main/resources to target/classes
[INFO]
```

```
[INFO] --- compiler:3.14.1:compile (default-compile) @ spring-petclinic ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 30 source files with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ spring-petclinic ---
[INFO] skip non existing resourceDirectory
/Users/dsarra/Downloads/Assignments/spring-petclinic/src/test/resources
[INFO]
[INFO] --- compiler:3.14.1:testCompile (default-testCompile) @ spring-petclinic ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 17 source files with javac [debug parameters release 17] to target/test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ spring-petclinic ---
[WARNING] Parameter 'systemProperties' is deprecated: Use systemPropertyVariables instead.
[INFO] Tests are skipped.
[INFO]
[INFO] --- jacoco:0.8.14:report (report) @ spring-petclinic ---
[INFO] Skipping JaCoCo execution due to missing execution data file.
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ spring-petclinic ---
[INFO] Artifact org.apache.maven.shared:file-management:pom:3.1.0 is present in the local repository, but cached
from a remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
[INFO] Artifact org.apache.maven.shared:file-management:pom:3.1.0 is present in the local repository, but cached
from a remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
Downloading from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/shared/file-management/3.1.0/file-manageme
nt-3.1.0.pom
Downloaded from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/shared/file-management/3.1.0/file-manageme
nt-3.1.0.pom (0 B at 0 B/s)
[INFO] Artifact org.apache.maven.shared:maven-shared-components:pom:36 is present in the local repository, but
cached from a remote repository ID that is unavailable in current build context, verifying that is downloadable from
[nexus (http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
[INFO] Artifact org.apache.maven.shared:maven-shared-components:pom:36 is present in the local repository, but
cached from a remote repository ID that is unavailable in current build context, verifying that is downloadable from
[nexus (http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
Downloading from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/shared/maven-shared-components/36/maven-
shared-components-36.pom
Downloaded from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/shared/maven-shared-components/36/maven-
shared-components-36.pom (0 B at 0 B/s)
[INFO] Artifact org.apache.maven:maven-parent:pom:36 is present in the local repository, but cached from a
remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
[INFO] Artifact org.apache.maven:maven-parent:pom:36 is present in the local repository, but cached from a
remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
Downloading from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/maven-parent/36/maven-parent-36.pom
Downloaded from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/maven-parent/36/maven-parent-36.pom (0 B at
0 B/s)
[INFO] Artifact org.apache.apache:pom:26 is present in the local repository, but cached from a remote repository ID
that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
```

```

[INFO] Artifact org.apache:apache:pom:26 is present in the local repository, but cached from a remote repository ID
that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
Downloading from nexus: http://localhost:8081/repository/maven-public/org/apache/apache/26/apache-26.pom
Downloaded from nexus: http://localhost:8081/repository/maven-public/org/apache/apache/26/apache-26.pom
(0 B at 0 B/s)
[INFO] Artifact org.apache.maven.shared:file-management:jar:3.1.0 is present in the local repository, but cached
from a remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
[INFO] Artifact org.apache.commons:commons-lang3:jar:3.14.0 is present in the local repository, but cached from
a remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
[INFO] Artifact org.apache.maven.shared:file-management:jar:3.1.0 is present in the local repository, but cached
from a remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
[INFO] Artifact org.apache.commons:commons-lang3:jar:3.14.0 is present in the local repository, but cached from
a remote repository ID that is unavailable in current build context, verifying that is downloadable from [nexus
(http://localhost:8081/repository/maven-public/, default, releases+snapshots)]
Downloading from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/shared/file-management/3.1.0/file-manageme
nt-3.1.0.jar
Downloading from nexus:
http://localhost:8081/repository/maven-public/org/apache/commons/commons-lang3/3.14.0/commons-lang3-3.
14.0.jar
Downloaded from nexus:
http://localhost:8081/repository/maven-public/org/apache/commons/commons-lang3/3.14.0/commons-lang3-3.
14.0.jar (0 B at 0 B/s)
Downloaded from nexus:
http://localhost:8081/repository/maven-public/org/apache/maven/shared/file-management/3.1.0/file-manageme
nt-3.1.0.jar (0 B at 0 B/s)
[INFO] Building jar:
/Users/dsarraaf/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:4.0.3:repackage (repackage) @ spring-petclinic ---
[INFO] Replacing main artifact
/Users/dsarraaf/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar with
repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to
/Users/dsarraaf/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar.original
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.926 s
[INFO] Finished at: 2026-05-31T11:33:24+05:30
[INFO] -----

```

b. Gradle Artifact Upload

Verify Gradle Installation: `./gradlew --version`

```
(.venv) dsarraafgriddynamics.com@C19548 spring-petclinic % ./gradlew --version
```

Downloading <https://services.gradle.org/distributions/gradle-9.2.1-bin.zip>
.....10%.....20%.....30%.....40%.....50%.....60%.....70%.....80%.....
90%.....100%

Welcome to Gradle 9.2.1!

Here are the highlights of this release:

- Windows ARM support
- Improved publishing APIs
- Better guidance for dependency verification failures

For more details see <https://docs.gradle.org/9.2.1/release-notes.html>

Gradle 9.2.1

Build time: 2025-11-17 13:40:48 UTC

Revision: 30ecdc708db275e8f8769ea0620f6dd919a58f76

Kotlin: 2.2.20

Groovy: 4.0.28

Ant: Apache Ant(TM) version 1.10.15 compiled on August 25 2024

Launcher JVM: 17.0.18 (Homebrew 17.0.18+0)

Daemon JVM: /opt/homebrew/Cellar/openjdk@17/17.0.18/libexec/openjdk.jdk/Contents/Home (no JDK specified, using current Java home)

OS: Mac OS X 26.5 aarch64

Publish Artifact - `./gradlew publish -x test`

```
● (.venv) rpavani@C19536 spring-petclinic % ./gradlew publish -x test
```

```
> Task :cyclonedxDirectBom
```

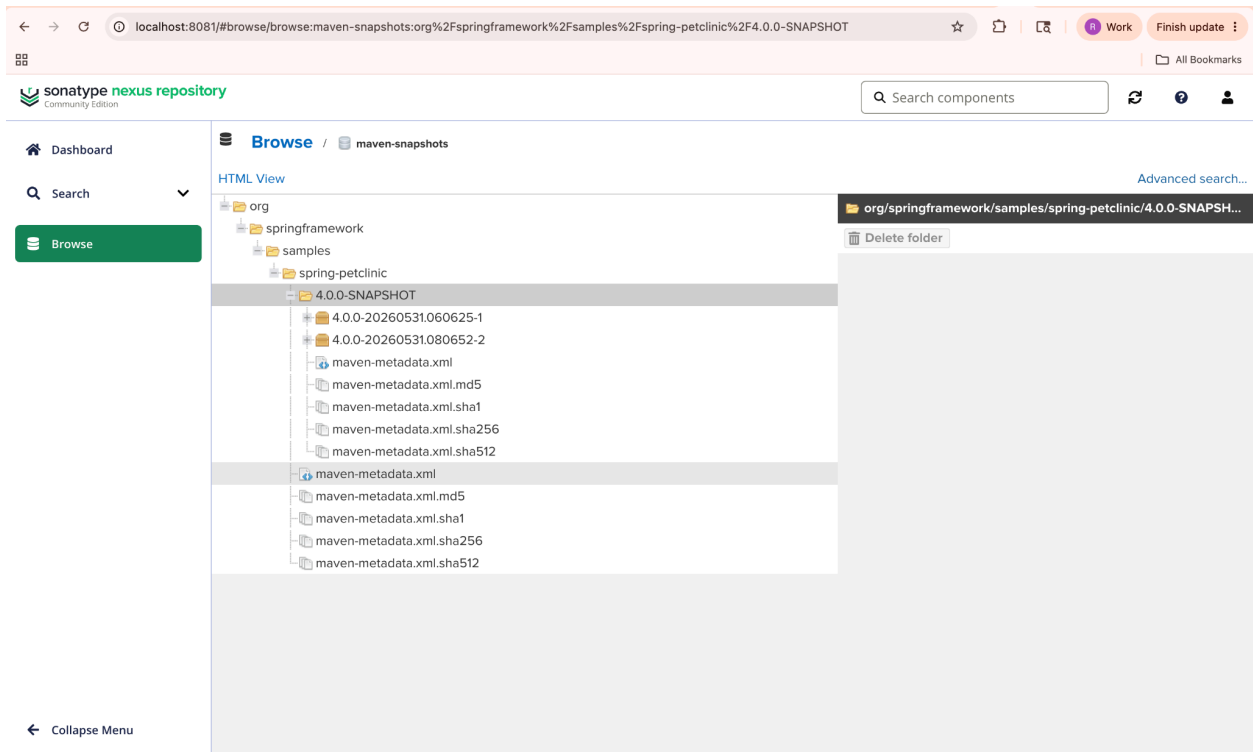
```
Unknown keyword meta:enum – you should define your own Meta Schema. If the keyword is irrelevant for validation, just use a NonValidationKeyword or if it should generate annotations AnnotationKeyword
```

```
Unknown keyword deprecated – you should define your own Meta Schema. If the keyword is irrelevant for validation, just use a NonValidationKeyword or if it should generate annotations AnnotationKeyword
```

```
BUILD SUCCESSFUL in 8s
```

```
8 actionable tasks: 8 executed
```

```
○ (.venv) rpavani@C19536 spring-petclinic % █
```



5. Upload Artifact to Nexus

The generated artifact was deployed to Nexus.

Command

```
./mvnw clean deploy -DskipTests
```

Deployment Target

<http://localhost:8081/repository/maven-snapshots/>

Result:

- Artifact uploaded successfully.
- Maven metadata generated automatically.
- Snapshot artifact stored in Nexus.

```

(.venv) rpavani@C19536 spring-petclinic % ./mvnw clean deploy -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.springframework.samples:spring-petclinic >-----
[INFO] Building petclinic 4.0.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ spring-petclinic ---
[INFO] Deleting /Users/rpavani/Downloads/Assignments/spring-petclinic/target
[INFO]
[INFO] --- enforcer:3.6.2:enforce (enforce-java) @ spring-petclinic ---
[INFO] Rule 0: org.apache.maven.enforcer.rules.version.RequireJavaVersion passed
[INFO]
[INFO] --- spring-javaformat:0.0.47:validate (default) @ spring-petclinic ---
[INFO]
[INFO] --- checkstyle:3.6.0:check (nohttp-checkstyle-validation) @ spring-petclinic ---
[INFO] You have 0 Checkstyle violations.
[INFO]
[INFO] --- jacoco:0.8.14:prepare-agent (default) @ spring-petclinic ---
[INFO] argLine set to -javaagent:/Users/rpavani/.m2/repository/org/jacoco/org.jacoco.agent/0.8.14/org.jacoco.agent-0.8.14-runtime.jar=destfile=/Users/rpavani/Downloads/Assignments/spring-petclinic/target/jacoco.exec
[INFO]
[INFO] --- git-commit-id:9.0.2:revision (default) @ spring-petclinic ---
[INFO] dotGitDirectory '/Users/rpavani/Downloads/Assignments/.git'
[INFO] Collected git.build.user.name with value rpavani
[INFO] Collected git.build.user.email with value rpavani@griddynamics.com
[INFO] Collected git.branch with value main
[INFO] Collected git.commit.id.describe with value d2b637c-dirty
[INFO] Collected git.commit.id.describe-short with value d2b637c-dirty
[INFO] Collected git.commit.id with value d2b637c687717cfeeb7b197a391ba6c087a53243
[INFO] Collected git.commit.id.abbrev with value d2b637c
[INFO] Collected git.dirty with value true
[INFO] Collected git.commit.user.name with value rpavani
[INFO] Collected git.commit.user.email with value rpavani@griddynamics.com
[INFO] Collected git.commit.message.full with value docker practical task
[INFO] Collected git.commit.message.short with value docker practical task
[INFO] Collected git.commit.time with value 2026-05-01T11:51:43+05:30
[INFO] Collected git.commit.author.time with value 2026-05-01T11:51:43+05:30
[INFO] Collected git.commit.committer.time with value 2026-05-01T11:51:43+05:30
[INFO] Collected git.remote.origin.url with value https://github.com/rpavani-png/Assignments.git
[INFO] Collected git.tags with value
[INFO] Collected git.tag with value
[INFO] Collected git.closest.tag.name with value
[INFO] Collected git.closest.tag.commit.count with value
[INFO] Collected git.total.commit.count with value 29
[INFO] Collected git.local.branch.ahead with value 0
[INFO] Collected git.local.branch.behind with value 0
[INFO] Collected git.build.time with value 2024-11-28T20:07:52+05:30
[INFO] Collected git.build.version with value 4.0.0-SNAPSHOT
[INFO] Collected git.build.host with value C19536
[INFO] including property 'git.tags' in results
[INFO] including property 'git.build.version' in results
[INFO] including property 'git.branch' in results
[INFO] including property 'git.build.host' in results
[INFO] including property 'git.commit.id' in results
[INFO] including property 'git.commit.user.email' in results

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[INFO] including property 'git.commit.id' in results
[INFO] including property 'git.commit.user.email' in results
[INFO] including property 'git.local.branch.behind' in results
[INFO] including property 'git.commit.author.time' in results
[INFO] including property 'git.build.user.name' in results
[INFO] including property 'git.dirty' in results
[INFO] including property 'git.closest.tag.commit.count' in results
[INFO] including property 'git.commit.user.name' in results
[INFO] including property 'git.commit.id.abbrev' in results
[INFO] including property 'git.commit.id.describe-short' in results
[INFO] including property 'git.total.commit.count' in results
[INFO] including property 'git.commit.id.describe' in results
[INFO] including property 'git.build.user.email' in results
[INFO] including property 'git.commit.message.short' in results
[INFO] including property 'git.commit.committer.time' in results
[INFO] including property 'git.tag' in results
[INFO] including property 'git.closest.tag.name' in results
[INFO] including property 'git.local.branch.ahead' in results
[INFO] including property 'git.commit.time' in results
[INFO] including property 'git.build.time' in results
[INFO] including property 'git.commit.message.full' in results
[INFO] including property 'git.remote.origin.url' in results
[INFO] Writing properties file [/Users/rpavani/Downloads/Assignments/spring-petclinic/target/classes/git.properties] (for project petclinic)...
[INFO]
[INFO] --- spring-boot:4.0.3:build-info (default) @ spring-petclinic ---
[INFO]
[INFO] --- cyclonedx:2.9.1:makeAggregateBom (default) @ spring-petclinic ---
[INFO] CycloneDX: Resolving Dependencies
[INFO] CycloneDX: Creating BOM version 1.6 with 108 component(s)
[INFO] CycloneDX: Writing and validating BOM (JSON): /Users/rpavani/Downloads/Assignments/spring-petclinic/target/classes/META-INF/sbom/application.cdx.json
[WARNING] Unknown keyword meta:enum - you should define your own Meta Schema. If the keyword is irrelevant for validation, just use a NonValidationKeyword or if it should generate annotations AnnotationKeyword
[WARNING] Unknown keyword deprecated - you should define your own Meta Schema. If the keyword is irrelevant for validation, just use a NonValidationKeyword or if it should generate annotations AnnotationKeyword
[INFO] attaching as spring-petclinic-4.0.0-SNAPSHOT-cyclonedx.json
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ spring-petclinic ---
[INFO] Copying 3 resources from src/main/resources to target/classes
[INFO] Copying 45 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.14.1:compile (default-compile) @ spring-petclinic ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 30 source files with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ spring-petclinic ---
[INFO] skip non existing resourceDirectory /Users/rpavani/Downloads/Assignments/spring-petclinic/src/test/resources
[INFO]
[INFO] --- compiler:3.14.1:testCompile (default-testCompile) @ spring-petclinic ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 17 source files with javac [debug parameters release 17] to target/test-classes
[INFO]

```



```

(.env) rpavani@C19536 spring-petclinic % ./mvnw clean deploy -DskipTests
[INFO] Building jar: /Users/rpavani/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:4.0.3:repackage (repackage) @ spring-petclinic ---
[INFO] Replacing main artifact /Users/rpavani/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to /Users/rpavani/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar.original
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ spring-petclinic ---
[INFO] Installing /Users/rpavani/Downloads/Assignments/spring-petclinic/pom.xml to /Users/rpavani/.m2/repository/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-SNAPSHOT.pom
[INFO] Installing /Users/rpavani/Downloads/Assignments/spring-petclinic/target/spring-petclinic-4.0.0-SNAPSHOT.jar to /Users/rpavani/.m2/repository/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-SNAPSHOT.jar
[INFO] Installing /Users/rpavani/Downloads/Assignments/spring-petclinic/target/classes/META-INF/sbom/application.cdx.json to /Users/rpavani/.m2/repository/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-SNAPSHOT-cyclonedx.json
[INFO]
[INFO] --- deploy:3.1.4:deploy (default-deploy) @ spring-petclinic ---
Downloading from nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/maven-metadata.xml
Uploading to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-20260531.060625-1.pom
Uploading to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-20260531.060625-1.jar
Uploading to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-20260531.060625-1-cyclonedx.json
Uploaded to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-20260531.060625-1.pom (15 kB at 387 kB/s)
Uploaded to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-20260531.060625-1-cyclonedx.json (278 kB at 7.1 MB/s)
Uploaded to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/spring-petclinic-4.0.0-20260531.060625-1.jar (69 MB at 76 MB/s)
Downloading from nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/maven-metadata.xml
Uploading to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/maven-metadata.xml
Uploaded to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/4.0.0-SNAPSHOT/maven-metadata.xml (1.0 kB at 46 kB/s)
Uploading to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/maven-metadata.xml
Uploaded to nexus: http://localhost:8081/repository/maven-snapshots/org/springframework/samples/spring-petclinic/maven-metadata.xml (303 B at 19 kB/s)
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 5.884 s
[INFO] Finished at: 2026-05-31T11:36:31+05:30
[INFO]

```


Search and Verify Artifact

Artifact verification was performed using the Nexus Browse functionality.

Repository Path

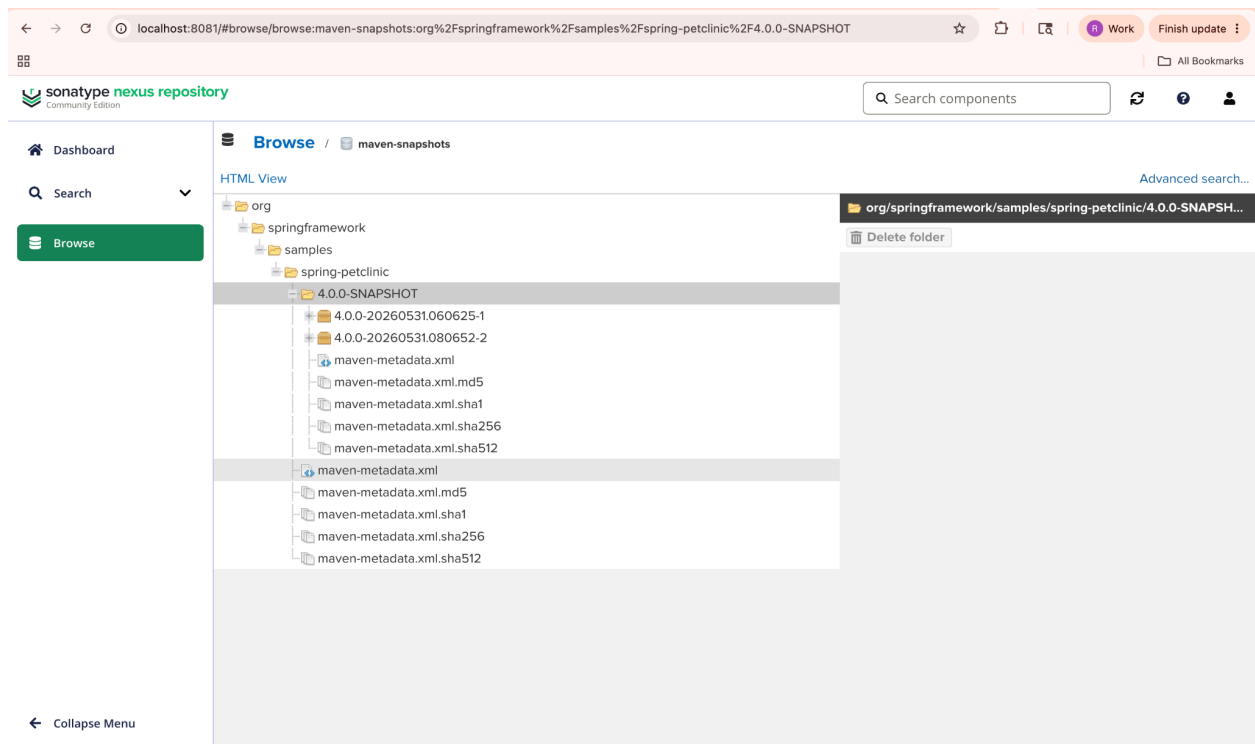
maven-snapshots

└─ org
 └─ springframework
 └─ samples
 └─ spring-petclinic
 └─ 4.0.0-SNAPSHOT

Verification

The following files were successfully uploaded:

- spring-petclinic-4.0.0-jar
- spring-petclinic-4.0.0-pom
- maven-metadata.xml



Challenges Encountered

Deployment Failure to maven-releases

Issue:

Repository version policy: RELEASE does not allow version 4.0.0-SNAPSHOT

Resolution:

- Updated distributionManagement to use the maven-snapshots repository.

Conclusion

Successfully installed and configured Sonatype Nexus Repository Manager. Maven was configured to use Nexus as a mirror repository, Spring Petclinic artifacts were built and deployed successfully to Nexus, and the uploaded artifacts were verified through the Nexus Browse interface.